

An Analysis of 2017-2023 Car Crashes in Brazil

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Introduction

A country's economy can often be reflected by the proportion of vehicles owned by its population. Those countries that have experienced economic growth tend to also observe an increase in urbanization, which generally leads to an increase in vehicle-to-person ratio (Sommer, Dargay & Gately 2007). In recent years, Brazil's economy has been experiencing growth, as real GDP and other overall macroeconomic indicators have shown improvement (World Bank). It would be reasonable to assume that if Brazil's economy continued growing as it has been recently, the number of vehicles on their roads would increase accordingly. As driving becomes more prevalent, car crashes would naturally become a larger concern, especially if the quality of the roads struggles to keep up with the influx of cars. This write-up examines an accumulation of reported vehicle crashes in Brazil from 2017 to 2023, with the intention of determining a resonable approach to ensure safety as a driver. In order to maintain structure in data analysis, crashes that result in death are used to demonstrate instances of a lack of safety.

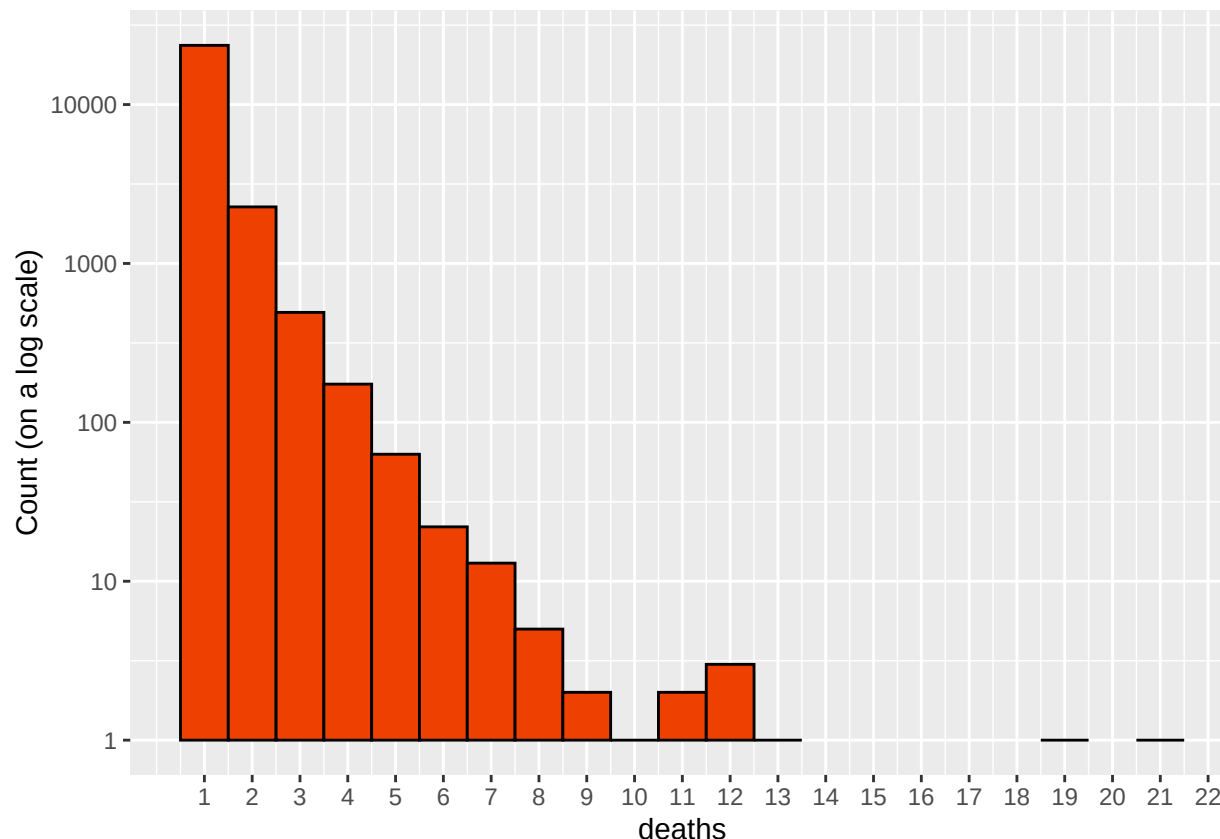
Summary of Data

The data set was accessed April 27th, 2025. It compromises of 463,152 data points in total, where each point represents the occurrence of a unique accident. The data was sourced from Kaggle (sent over by Prof. O'Brien), and was compiled from several datasets created by Brazil's Federal Highway Police Data, reported on the government's Ministry of Justice and Public Service website.

Each of the accidents provides us with a plethora of information. The focus of this analysis, however, revolves around a limited number of features (dimensions) deemed to be potentially influential to the occurrence of the accident. Those features are compromised of the cause of the accident (which intuitively seems like the most related to fatality), the weather, the time of day, day of the week, and the road delineation at the scene. In order to properly synthesize these features, it was important to clean the data appropriately by removing any blank data points from the data set, as well as identifying any uncertain feature values, such as points with road delineation values listed as "Not Reported". A full list of features and a link to the dataset on Kaggle are listed below in the appendix.

Visualizing Deaths

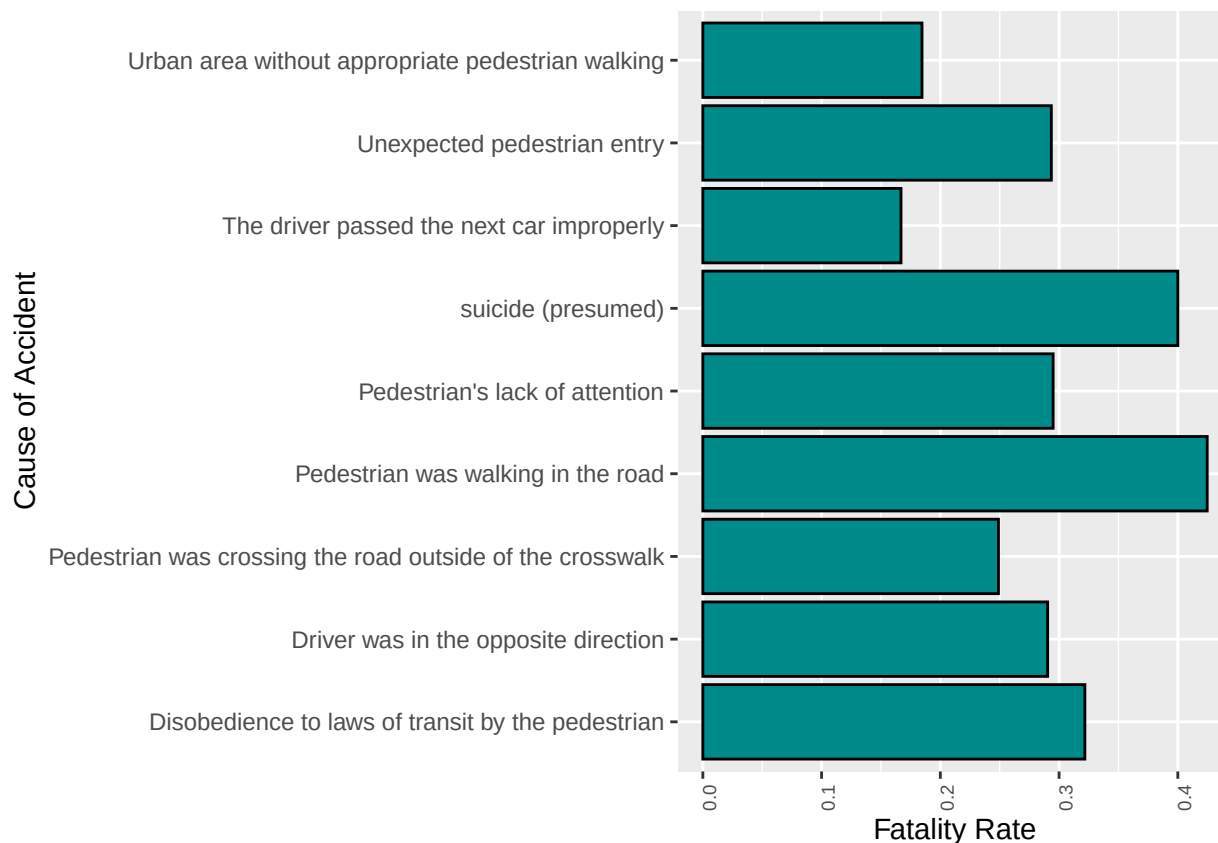
Using a bar graph provides a straightforward way to understand how the number of deaths are distributed for fatal accidents.



It was clear that the vast majority of said accidents only resulted in one death; finding accidents with a certain number of deaths becomes increasingly harder to find, until reaching 10 deaths. Log-scaling the y-axis allows for a cleaner visualization here.

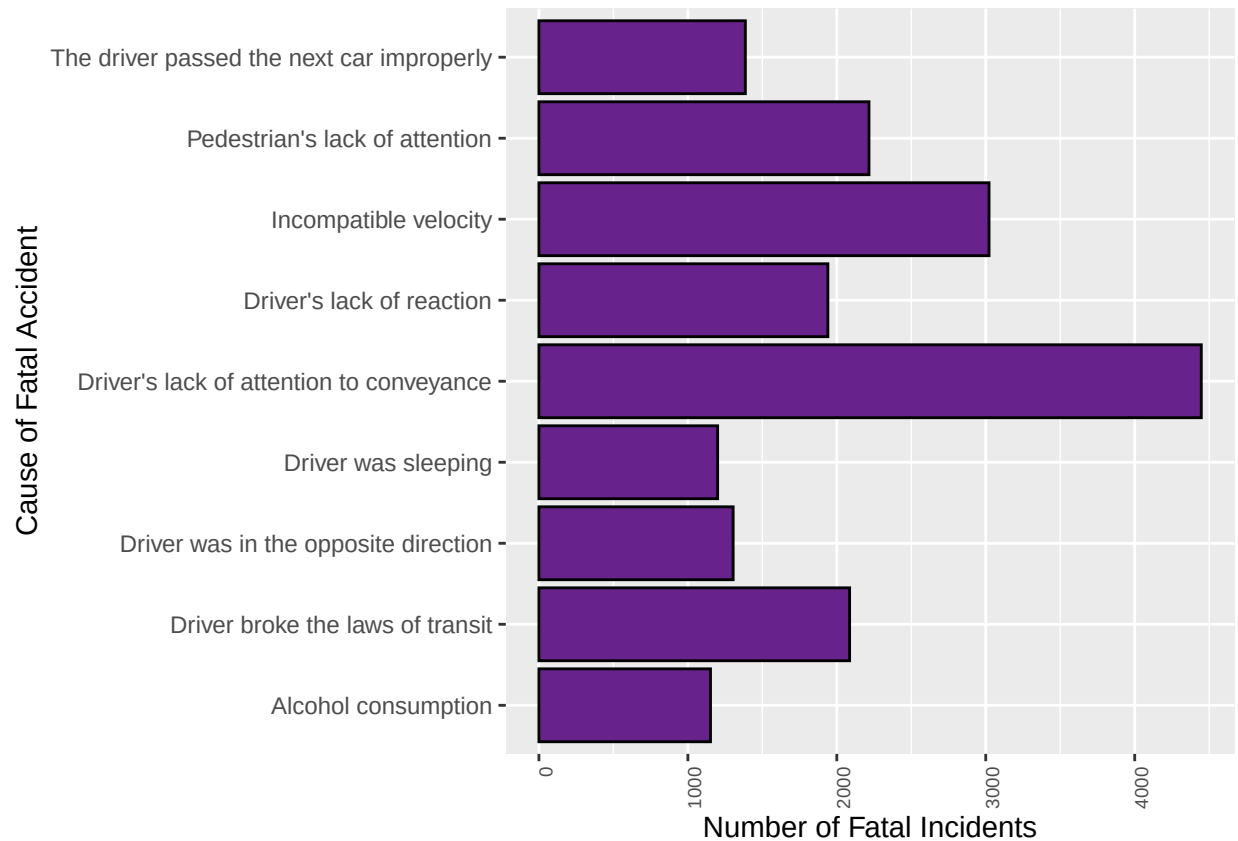
It becomes a lot harder to put a number on how many fatalities a single accident will cause, given that there are at least 10. Many of these can be attributed to accidents involving special types of vehicles such as buses or trucks. For example, the accident that killed 19 people was a bus collision in Guaratuba in 2021, and was related to a brake malfunction (Farmbauer 2021). Including these types of accidents (involving 10+ deaths) in the analysis comes with its own risks, as the circumstances for accidents involving larger vehicles can be unique. However, they occur so infrequently, and could potentially be caused by the fault of car-drivers and pedestrians (which cause a ton of smaller-scale accidents as will be shown later), so excluding them is not entirely necessary.

Determining how severe crashes have to be to function as significant indicators of a lack of safety is not easy. Deciding on how much harm an accident has to cause to warrant attention is subjective, as there is no true baseline for being a safe driver, unless every single minute of every driver being unharmed is considered. In order to conduct an analysis that isn't overwhelming to interpret, the decision was made to categorize all accidents into fatal or not fatal. Although this is already covered in the victims' condition feature, creating a new binary representation of whether or not a death occurred warrants readability in the results.



The above bar graph examines the causes of accidents that lead to the highest probability of at least one death occurring ($P(\text{fatal} \mid \text{cause of accident})$). One can note that the causes with most of 9 causes with the highest fatality rates involve pedestrians. It can logically be presumed that this is the case due to the lack of physical protection pedestrians have whenever they are involved in accidents, also begging the question of what exactly constitutes of an accident occurring because of a “pedestrian walking in the road”, or a “pedestrian’s lack of attention”.

Additionally, suicide is shown to have a fatality rate of approximately 0.4, although it is worth noting that the dataset only contains 5 instances of suicide.



The above data gives a more holistic perspective of what to expect as the reasons for most fatal accidents. If a fatal accident were to occur right now, it is very likely that one of these causes would be the culprit.

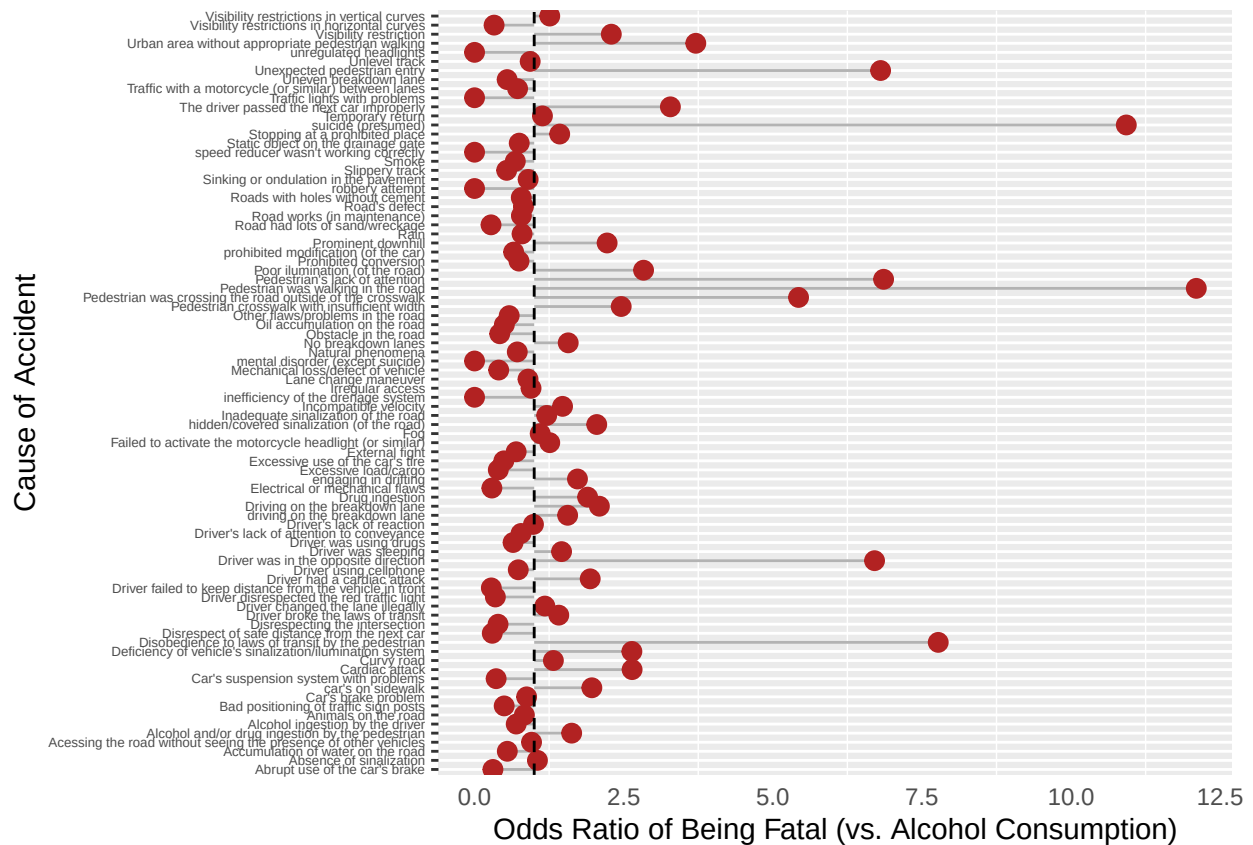
```
intersect(df.1[1], df.2[1])
```

```
## # A tibble: 3 x 1
##   cause_of_accident
##   <chr>
## 1 Driver was in the opposite direction
## 2 Pedestrian's lack of attention
## 3 The driver passed the next car improperly
```

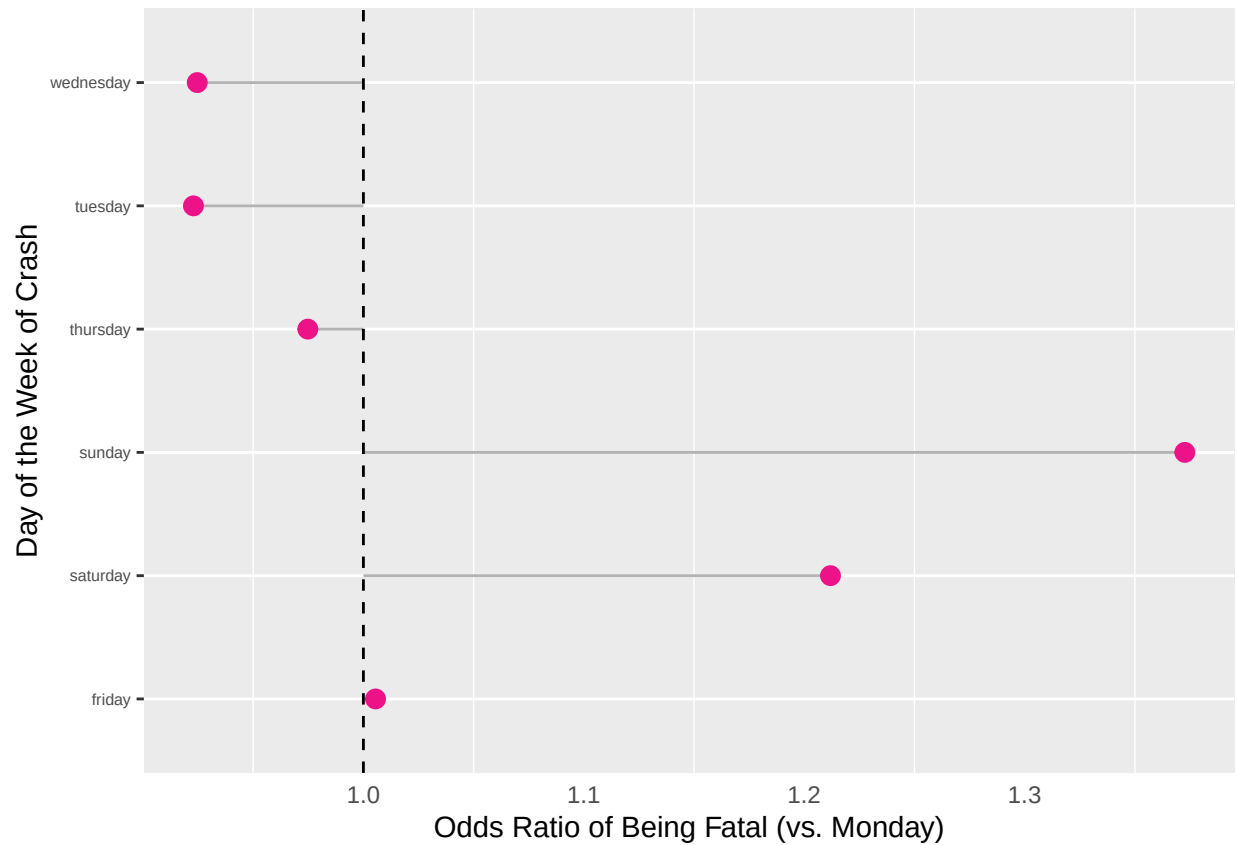
These three are the causes of fatal accidents that appear in both subsets of data, making them reasonable to prioritize when thinking about safety.

Models

In order to better understand what kind of events have the highest fatality rates, we can use logistic regression (since there are only two outcomes, fatal or not fatal). Although logistic regression works well for a binary classification, we are using categorical variables as our predictors (e.g. weather condition), which means that a reference variable (e.g. rainy) has to be chosen in order for regression to make sense.

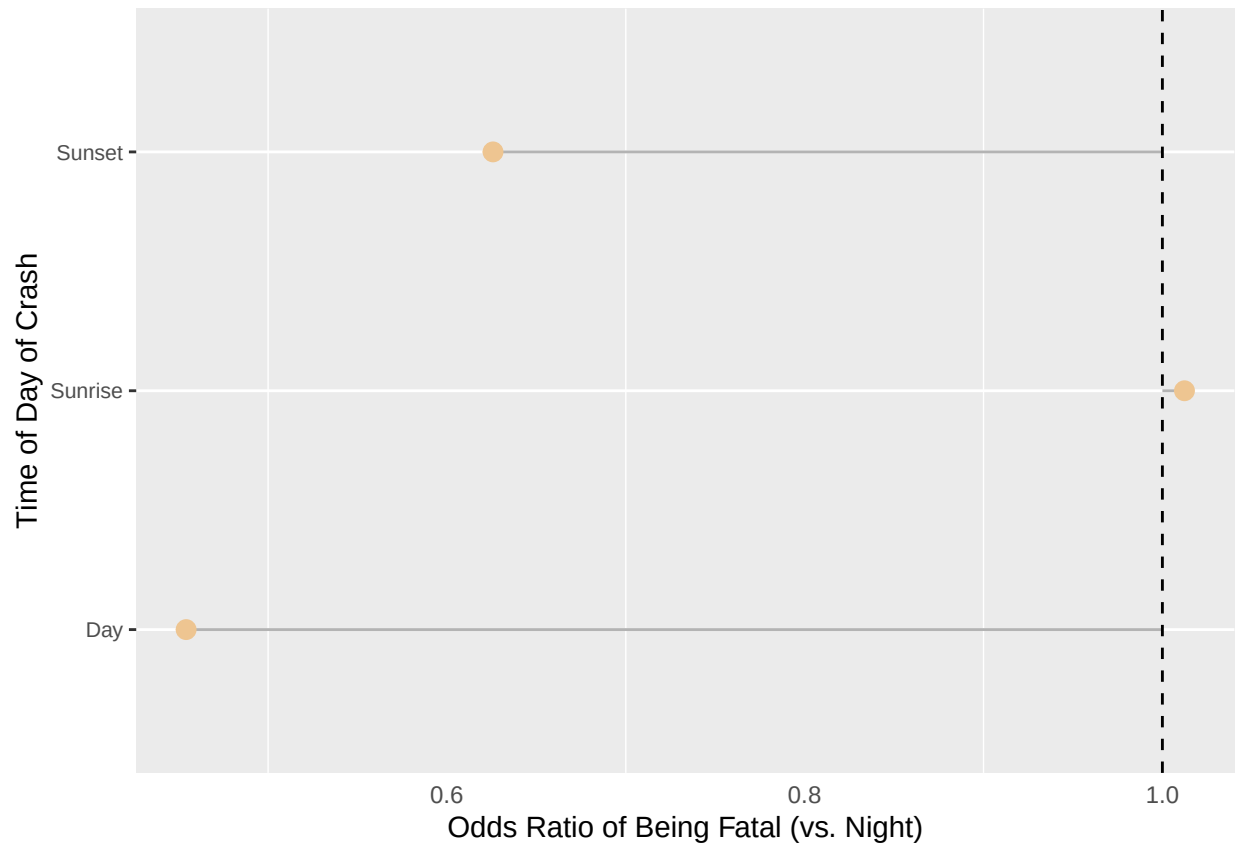


Here are the odds ratios of causes of an accidents being fatal when compared to accidents caused by alcohol consumption. As observed previously in our bar graph of types of accident causes by fatality rate, pedestrian accidents tend to be more fatal. Interestingly enough, accidents involving drivers using cellphones have much lower odds of being fatal, and accidents involving curvy roads have slightly higher odds of being fatal. Poor illumination of the road and urban areas without appropriate pedestrian walking also appear to have slightly higher odds of being fatal. More importantly, the odds of fatal accidents involved with pedestrians walking in the road is much higher than the odds of fatal accidents involved with alcohol consumption (ratio of 12). Although this has been picked up on as an issue in our earlier bar graph, there now might be some context behind it; the fact that being in an accident involving walking on a poorly-planned road for pedestrians is associated with death implies that fatalities regarding people walking in the middle of the road might have to do with the country's walkable infrastructure.

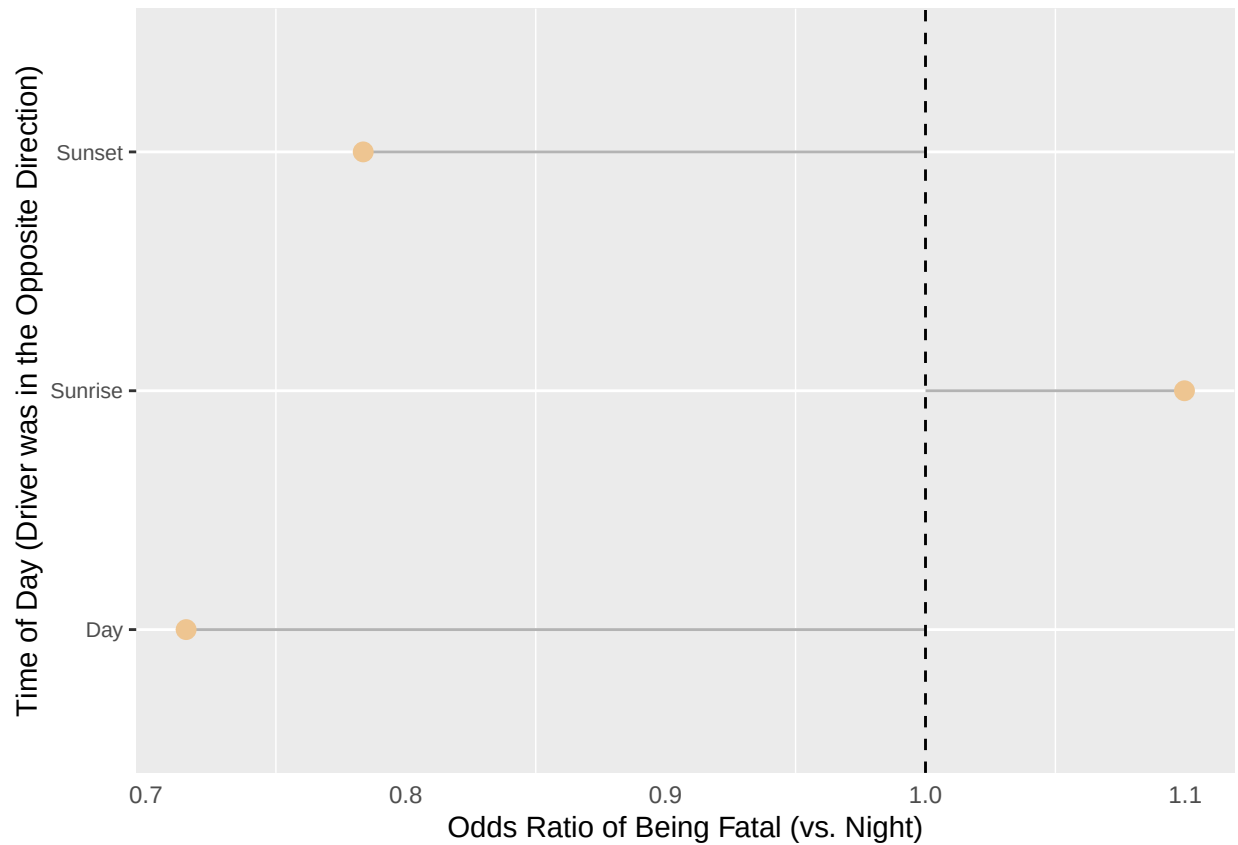


Examining the odds ratio of how fatal accidents are during the days of the week provides a completely new perspective, since they are a classification of time where each event (day) occurs evenly. This differs from the causes of the accidents, which do not share the same uniformity.

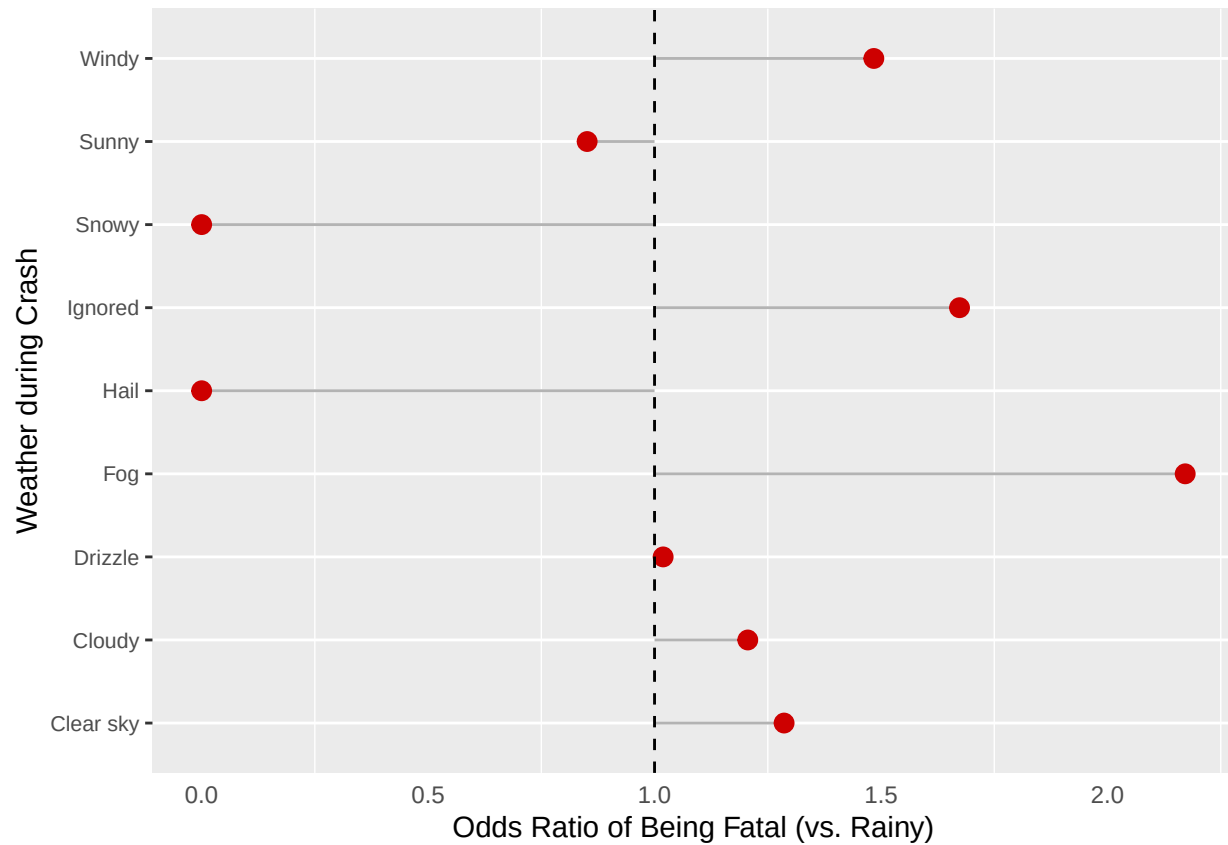
Interestingly enough, when examining the odds ratio of being fatal compared to Mondays, crashes that happen on Sunday have the highest ratio (almost 1.4), however these ratios don't have nearly the same magnitude as those from the cause of accident feature did. Tuesdays, Wednesdays, and Thursdays appear to have odds ratios slightly lower than 1.



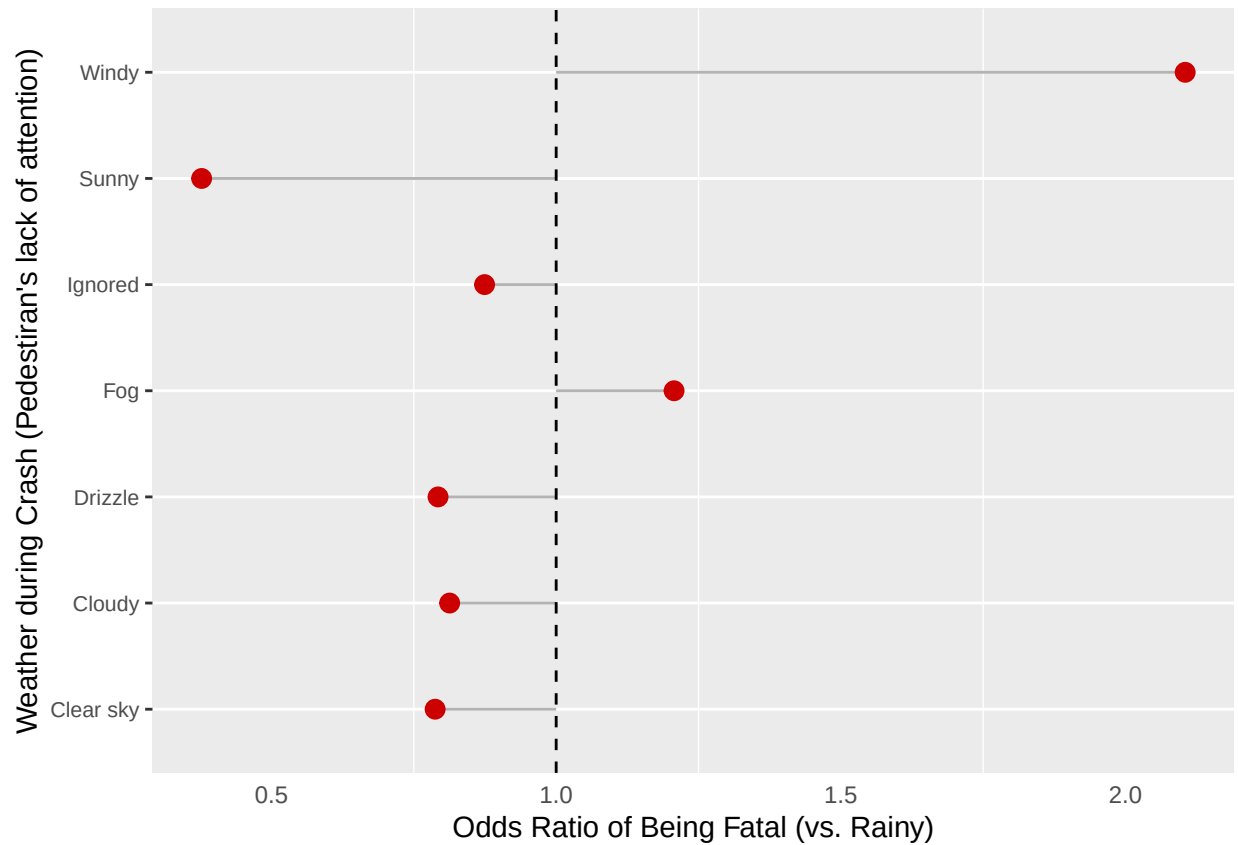
Considering the odds of accidents being fatal regarding time of day also benefits from the same uniformity of time that days of the week exhibited. Using nighttime as a reference for our odds ratios, the above graph displays how much higher the odds of causing death each of the other times of day are. surprisingly enough, sunrise accidents appear to be a bit more likely to cause death, while daytime accidents are way less likely to cause death. This could be attributed to traffic levels in the morning as people are going to work, and the susceptibility of road rage as people are adjusting from waking up.



Only considering accidents involving drivers going in the wrong direction (a common cause of death) for the graph above, it is evident that this increases the odds ratio of being fatal significantly during sunrise as well as during sunset and daytime. This suggests that driving at night wouldn't necessarily increase one's chances of dying from a wrong way driver when compared to driving during the daytime or sunset, but driving at sunrise could definitely make this more of a risk. Perhaps this kind of accident is related to levels of traffic.



It is also helpful to consider odds ratios of types of weather conditions in causing crashes. Rain is used as a reference, since hydroplaning could contribute to a loss of control on the road. Interestingly enough, snow has a smaller odds ratio. This probably has to do with how rare it is in a country like Brazil, contributing to extra precaution by the population when it does happen. Fog has the highest odds ratio at ~2.2, which makes sense as it reduces visibility.



Above, only accidents involving pedestrians' lack of attention are considered. There is a clear indication of wind being the most risky to pedestrians, as it has the highest odds ratio. One could infer that too much wind poses a great distraction for pedestrians, as it can cause eye irritation which can make it harder for them to respond to road situations appropriately.



It is clear by this diagram that tunnels, roundabouts, and overpasses are all relatively safe compared to curves. Surprisingly, intersections are also relatively safe in terms of fatal accidents. Although overpasses tend to be safer than curves, bridges tend to be much more dangerous. This calls into question the quality of the infrastructure of Brazilian bridges. Is it too easy for cars to fall into water? Or maybe this is something specific to bridges in general, due to drivers getting anxious as they try to cross, impairing their driving abilities.

```
##           df      AIC
## vs.coa.model      85 181309.0
## vs.weekday.model    7 195732.5
## vs.daytime.model    4 192633.0
## vs.weather.model   10 195784.1
## vs.delineation.model 9 194805.3
```

Intuitively, the cause of the accident is the most important predictor in whether it is fatal or not, since hitting someone's bumper is a lot less likely to kill someone than running into a pedestrian. Here we can see that the feature for the cause has the lowest AIC, implying the use of it as a predictor as superior. Interestingly enough, time of day and road delineation function as better predictors than day of the week and weather condition (but not by much).

```
##      df      AIC
## model11 88 178950.0
## model12 94 181044.7
## model13 97 178810.0
## model14 96 177746.5
## model15 105 177592.9
```

In the above table, model1 is the result of using both the cause of accident and the time of day to predict fatalities, model2 is the result of using the cause of accident and the weather condition, model3 is from using the cause of accident with both the time of day and the weather condition, and model4 is using the cause of accident, time of day, and road delineation. Because the days of the week didn't end up having that much variance in odds ratios, it didn't seem worth including.

It is hard to determine the predictors with the best fit, however an effort can be made to choose those that explain different parts of why a fatal accident occurred (for example, maybe the cause of the accident wouldn't have caused as severe of a crash if it happened during the day instead of during sunrise).

Adding weather condition as a second predictor to cause of accident seems to worsen the regression model (higher AIC). In spite of this, using cause of accident, weather condition, and time of day as the three predictors works better than just using cause of accident with time of day. This implies that the three of these features explain different factors to why an accident ends up being fatal. We can also observe that replacing weather condition with road delineation has the lowest AIC, making it more useful than using all four predictors. This makes model4 the most accurate model tested.

Results

In order to fully reach a conclusion, we must analyze both the odds ratios of the coefficients of our predictor features as well as conduct some hypothesis testing on them.

##	Predictor	Estimate
## 1	Driver was in the opposite direction	1.9632305
## 2	Pedestrian was walking in the road	2.4422606
## 3	Pedestrian's lack of attention	1.9845367
## 4	Day	-0.6628993
## 5	The driver passed the next car improperly	1.3698386
## 6	Unexpected pedestrian entry	1.9937484
## 7	Disobedience to laws of transit by the pedestrian	2.0955628
## 8	Pedestrian was crossing the road outside of the crosswalk	1.7636236
## 9	Intersection	-1.0067748
## 10	Cardiac attack	1.2224545
## 11	Driver broke the laws of transit	0.6616434
## 12	Roundabout	-1.2488525
## 13	Incompatible velocity	0.5502768
## 14	Disrespect of safe distance from the next car	-0.9605678
## 15	Sunset	-0.4088906
## 16	Driver failed to keep distance from the vehicle in front	-1.0303039
## 17	Visibility restriction	0.8633556
## 18	Mechanical loss/defect of vehicle	-0.7295840
## 19	Authorized U-turn	-0.9702914
## 20	Electrical or mechanical flaws	-1.0778269
## 21	Driver had a cardiac attack	0.9128295
## 22	Deficiency of vehicle's sinalization/ilumination system	0.8778932
##	Odds.Ratio	p.value
## 1	7.122	0.00e+00
## 2	11.499	0.00e+00
## 3	7.276	0.00e+00
## 4	0.515	0.00e+00
## 5	3.935	2.79e-224
## 6	7.343	2.63e-209
## 7	8.130	6.72e-168
## 8	5.834	3.50e-141

## 9	0.365	1.80e-95
## 10	3.396	3.11e-80
## 11	1.938	3.47e-66
## 12	0.287	3.09e-58
## 13	1.734	9.12e-51
## 14	0.383	8.22e-46
## 15	0.664	1.71e-41
## 16	0.357	4.65e-36
## 17	2.371	1.35e-31
## 18	0.482	1.02e-30
## 19	0.379	1.09e-27
## 20	0.340	1.65e-25
## 21	2.491	3.70e-21
## 22	2.406	6.20e-21

To test whether the coefficients in the features we are interested in hold significance in predicting the occurrence of deaths, I used R's `summary()` function. The p-values obtained through this method indicate whether or not there's enough evidence to suggest whether or not the null hypothesis that that specific predictor coefficient is not influencing the outcome of deaths, is wrong.

As shown in the graph above, there are a few predictor coefficients that have significantly low p values. As expected, pedestrian's lack of attention is a great predictor of whether an accident will end up being fatal. The same thing goes with pedestrians walking in the road. Drivers in the opposite direction as well as drivers passing the next car improperly also made the cut, as well as a few other pedestrian-related occurrences. However daytime and intersection are both up there in terms of having low p-values, which implies that they are statistically significant, however they have low odds ratios and negative estimates, meaning they are good predictors of crashes not being fatal.

Conclusion

This project analyzed the reasons behind fatal accidents, as a way to help infer how to avoid them. Issues caused by pedestrians were the most common top indicators of a fatal car crash, implying a mixture between not being protected by a vehicle of their own and a lack of suitable infrastructure for those who choose to walk instead of drive. It was also recognized that drivers going in the opposite direction strongly indicated a fatal car crash, implying a mixture of the impact of going in opposite directions alongside a possible lack of suitable indicators for drivers, such as signs, and a possible lack of respect for the law by the drivers. These results were not just statistically significant, but they occurred frequently enough to justify them being considerable issues for safety. This indicates that as Brazil becomes more urbanized, a higher investment in road infrastructure is necessary.

Appendix

Data Source / Features <https://www.kaggle.com/datasets/mlipppo/car-accidents-in-brazil-2017-2023/data>

inverse_data, week_day, hour, state, road_id, km, city, cause_of_accident, type_of_accident, victims_condition, weather_timestamp, road_direction, weather_condition, road_type, road_delineation, people, deaths, slightly_injured, severely_injured, uninjured, ignored, total_injured, vehicles_involved, latitude, longitude, regional, police_station.

References Sommer, Martin, Joyce Dargay, and Dermot Gately. "Vehicle Ownership and Income Growth, Worldwide: 1960–2030." *The Energy Journal* 28, no. 4 (2007): 143–170.

“The World Bank In Brazil.” World Bank, April 30, 2025.

Farmbauer, Martina. “At Least 19 Killed in Brazil Bus Crash.” The Senior, January 25, 2021.

```
library(ggplot2)
library(corrplot)
library(dplyr)
library(tibble)

ds = read.csv("accidents.csv")
ds = na.omit(ds)

#drop crashes with unreported road delineation
ds = ds[ds$road_delineation != "Not Reported",]

names(ds)[names(ds)=="wheather_condition"] = "weather_condition"

ds.nonzero = ds[ds$deaths > 0,]

#asked chatgpt "how do i turn the y axis of my histogram into log scale r"
# and "how do i add better x axis tick marks" and "how do i add a border around each bar"

x = ggplot(ds.nonzero, aes(x = deaths)) +
  geom_histogram(binwidth = 1, fill = "orangered2", color = "black") +
  scale_y_log10() +
  scale_x_continuous(breaks = 1:40) +
  labs(y = "Log Count")

print(x)

#asked chatgpt "create a vector that stores each cause of accident as columns and its probability of be

ds$fatal = ifelse(ds$vicims_condition == "With dead victims", 1, 0)

cause.probs = ds %>%
  group_by(cause_of_accident) %>%
  summarise(fatality_rate = mean(fatal))

df.1 = cause.probs[cause.probs$fatality_rate >= .15,]

x = ggplot(df.1, aes(x = cause_of_accident, y = fatality_rate)) + geom_bar(stat = "identity", fill = "c
  scale_fill_hue(c = 40) +
  #asked chatgpt "how do i make x-axis labels more readable"
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 7)) +
  coord_flip() +
  labs(x = "Cause of Accident", y = "Fatality Rate")

print(x)

cause.freq = ds %>%
  group_by(cause_of_accident) %>%
  summarise(freq = sum(fatal))

df.2 = cause.freq[cause.freq$freq >= 1000,]
```

```

x = ggplot(df.2, aes(x = cause_of_accident, y = freq)) + geom_bar(stat = "identity", fill = "darkorchid",
scale_fill_hue(c = 40) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 7)) +
  coord_flip() +
  labs(x = "Cause of Fatal Accident", y = "Number of Fatal Incidents")

print(x)

intersect(df.1[1], df.2[1])

## Asked chatgpt: "help me get a glm logistic regression model whose coefficients are the log odds of A

ds$cause_of_accident = relevel(as.factor(ds$cause_of_accident), ref = "Alcohol consumption")

model = glm(fatal ~ cause_of_accident, data = ds, family = "binomial")

odds.ratios = exp(coef(model))

#asked chatgpt "how do i turn odds.ratios into a dataframe"
df.odds.r = data.frame(cause = names(odds.ratios), odds.ratios = as.numeric(odds.ratios))

df.odds.r = df.odds.r[df.odds.r$cause != "(Intercept)",]
df.odds.r$cause = gsub("cause_of_accident", "", df.odds.r$cause)

x = ggplot(df.odds.r, aes(x = cause, y = odds.ratios)) +
  geom_segment(aes(x = cause, xend = cause, y = 1, yend = odds.ratios), color = "gray70") +
  geom_point(color = "firebrick", size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "black") +
  coord_flip() +
  theme(
    panel.grid.major.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.text.y = element_text(size = 5)
  ) +
  xlab("Cause of Accident") +
  ylab("Odds Ratio of Being Fatal (vs. Alcohol Consumption)")
print(x)

ds$week_day = relevel(as.factor(ds$week_day), ref = "monday")

model = glm(fatal ~ week_day, data = ds, family = "binomial")

odds.ratios = exp(coef(model))

df.odds.r = data.frame(cause = names(odds.ratios), odds.ratios = as.numeric(odds.ratios))

df.odds.r = df.odds.r[df.odds.r$cause != "(Intercept)",]
df.odds.r$cause = gsub("week_day", "", df.odds.r$cause)

x = ggplot(df.odds.r, aes(x = cause, y = odds.ratios)) +
  geom_segment(aes(x = cause, xend = cause, y = 1, yend = odds.ratios), color = "gray70") +

```

```

geom_point(color = "deeppink2", size = 3) +
geom_hline(yintercept = 1, linetype = "dashed", color = "black") +
coord_flip() +
theme(
  panel.grid.major.x = element_blank(),
  axis.ticks.x = element_blank(),
  axis.text.y = element_text(size = 6)
) +
xlab("Day of the Week of Crash") +
ylab("Odds Ratio of Being Fatal (vs. Monday)")
print(x)

ds$weather_timestamp = relevel(as.factor(ds$weather_timestamp), ref = "Night")

model = glm(fatal ~ weather_timestamp, data = ds, family = "binomial")

odds.ratios = exp(coef(model))

df.odds.r = data.frame(cause = names(odds.ratios), odds.ratios = as.numeric(odds.ratios))

df.odds.r = df.odds.r[df.odds.r$cause != "(Intercept)",]
df.odds.r$cause = gsub("weather_timestamp", "", df.odds.r$cause)

x = ggplot(df.odds.r, aes(x = cause, y = odds.ratios)) +
  geom_segment(aes(x = cause, xend = cause, y = 1, yend = odds.ratios), color = "gray70") +
  geom_point(color = "burlywood2", size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "black") +
  coord_flip() +
  theme(
    panel.grid.major.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.text.y = element_text(size = 8)
  ) +
  xlab("Time of Day of Crash") +
  ylab("Odds Ratio of Being Fatal (vs. Night)")
print(x)

ds.wrong.way = ds[ds$cause_of_accident == "Driver was in the opposite direction",]

ds.wrong.way$weather_timestamp = relevel(as.factor(ds.wrong.way$weather_timestamp), ref = "Night")

model = glm(fatal ~ weather_timestamp, data = ds.wrong.way, family = "binomial")

odds.ratios = exp(coef(model))

df.odds.r = data.frame(cause = names(odds.ratios), odds.ratios = as.numeric(odds.ratios))

df.odds.r = df.odds.r[df.odds.r$cause != "(Intercept)",]
df.odds.r$cause = gsub("weather_timestamp", "", df.odds.r$cause)

x = ggplot(df.odds.r, aes(x = cause, y = odds.ratios)) +

```

```

geom_segment(aes(x = cause, xend = cause, y = 1, yend = odds.ratios), color = "gray70") +
geom_point(color = "burlywood2", size = 3) +
geom_hline(yintercept = 1, linetype = "dashed", color = "black") +
coord_flip() +
theme(
  panel.grid.major.x = element_blank(),
  axis.ticks.x = element_blank(),
  axis.text.y = element_text(size = 8)
) +
xlab("Time of Day (Driver was in the Opposite Direction)") +
ylab("Odds Ratio of Being Fatal (vs. Night)")
print(x)

ds$weather_condition = relevel(as.factor(ds$weather_condition), ref = "Rainy")

model = glm(fatal ~ weather_condition, data = ds, family = "binomial")

odds.ratios = exp(coef(model))

df.odds.r = data.frame(cause = names(odds.ratios), odds.ratios = as.numeric(odds.ratios))

df.odds.r = df.odds.r[df.odds.r$cause != "(Intercept)",]
df.odds.r$cause = gsub("weather_condition", "", df.odds.r$cause)

x = ggplot(df.odds.r, aes(x = cause, y = odds.ratios)) +
  geom_segment(aes(x = cause, xend = cause, y = 1, yend = odds.ratios), color = "gray70") +
  geom_point(color = "red3", size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "black") +
  coord_flip() +
  theme(
    panel.grid.major.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.text.y = element_text(size = 8)
  ) +
  xlab("Weather during Crash") +
  ylab("Odds Ratio of Being Fatal (vs. Rainy)")
print(x)

ds.pedest = ds[ds$cause_of_accident == "Pedestrian's lack of attention",]

ds.pedest$weather_condition = relevel(as.factor(ds.pedest$weather_condition), ref = "Rainy")

model = glm(fatal ~ weather_condition, data = ds.pedest, family = "binomial")

odds.ratios = exp(coef(model))

df.odds.r = data.frame(cause = names(odds.ratios), odds.ratios = as.numeric(odds.ratios))

df.odds.r = df.odds.r[df.odds.r$cause != "(Intercept)",]
df.odds.r$cause = gsub("weather_condition", "", df.odds.r$cause)

```

```

x = ggplot(df.odds.r, aes(x = cause, y = odds.ratios)) +
  geom_segment(aes(x = cause, xend = cause, y = 1, yend = odds.ratios), color = "gray70") +
  geom_point(color = "red3", size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "black") +
  coord_flip() +
  theme(
    panel.grid.major.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.text.y = element_text(size = 8)
  ) +
  xlab("Weather during Crash (Pedestrian's lack of attention)") +
  ylab("Odds Ratio of Being Fatal (vs. Rainy)")
print(x)

ds$road_delineation = relevel(as.factor(ds$road_delineation), ref = "Curve")

model = glm(fatal ~ road_delineation, data = ds, family = "binomial")

odds.ratios = exp(coef(model))

df.odds.r = data.frame(cause = names(odds.ratios), odds.ratios = as.numeric(odds.ratios))

df.odds.r = df.odds.r[df.odds.r$cause != "(Intercept)",]
df.odds.r$cause = gsub("road_delineation", "", df.odds.r$cause)

x = ggplot(df.odds.r, aes(x = cause, y = odds.ratios)) +
  geom_segment(aes(x = cause, xend = cause, y = 1, yend = odds.ratios), color = "gray70") +
  geom_point(color = "yellow4", size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "black") +
  coord_flip() +
  theme(
    panel.grid.major.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.text.y = element_text(size = 8)
  ) +
  xlab("Road Delineation during Crash") +
  ylab("Odds Ratio of Being Fatal (vs. Curve)")
print(x)

##asked chatgpt: "Is there a built-in way to calculate AIC?"

vs.coa.model = glm(fatal ~ cause_of_accident, data = ds, family = "binomial")

vs.weekday.model = glm(fatal ~ week_day, data = ds, family = "binomial")

vs.daytime.model = glm(fatal ~ weather_timestamp, data = ds, family = "binomial")

vs.weather.model = glm(fatal ~ weather_condition, data = ds, family = "binomial")

vs.delineation.model = glm(fatal ~ road_delineation, data = ds, family = "binomial")

aic = AIC(vs.coa.model, vs.weekday.model, vs.daytime.model, vs.weather.model, vs.delineation.model)

```

```

aic_df = as.data.frame(aic)

print(aic)

model1 = glm(fatal ~ cause_of_accident + weather_timestamp, data = ds, family = "binomial")
model2 = glm(fatal ~ cause_of_accident + weather_condition, data = ds, family = "binomial")
model3 = glm(fatal ~ cause_of_accident + weather_timestamp + weather_condition, data = ds, family = "binomial")
model4 = glm(fatal ~ cause_of_accident + weather_timestamp + road_delineation, data = ds, family = "binomial")
model5 = glm(fatal ~ cause_of_accident + weather_timestamp + road_delineation + weather_condition, data = ds, family = "binomial")

aic = AIC(model1, model2, model3, model4, model5)

print(aic)

##asked chatgpt "is there a built in way to get the p values of the null hypothesis of the categories"

summary_coefs = summary(model4)$coefficients

significant = summary_coefs[summary_coefs[, 4] < 1e-20, ]

significant_df = as.data.frame(significant)

significant_df = significant_df[rownames(significant_df) != "(Intercept)", ]

significant_df = significant_df[order(significant_df[, 4]), ]

term_names = rownames(significant_df)
term_names = gsub("cause_of_accident", "", term_names)
term_names = gsub("road_delineation", "", term_names)
term_names = gsub("weather_timestamp", "", term_names)
rownames(significant_df) = term_names

# Build results table using the cleaned names
results_df = data.frame(
  Predictor = term_names,
  Estimate = significant_df[, 1],
  `Odds Ratio` = round(exp(significant_df[, 1]), 3),
  `p-value` = format(significant_df[, 4], scientific = TRUE, digits = 3),
  stringsAsFactors = FALSE
)

print(results_df)

```